ISMRM 2025
MR Artifacts Game Show

Classic Artifacts in Radial MRI and The Semi-convergence Behavior of CG-SENSE

Hung Do, PhD MSEE
Canon Medical Systems USA

Declaration of
Financial Interests or Relationships

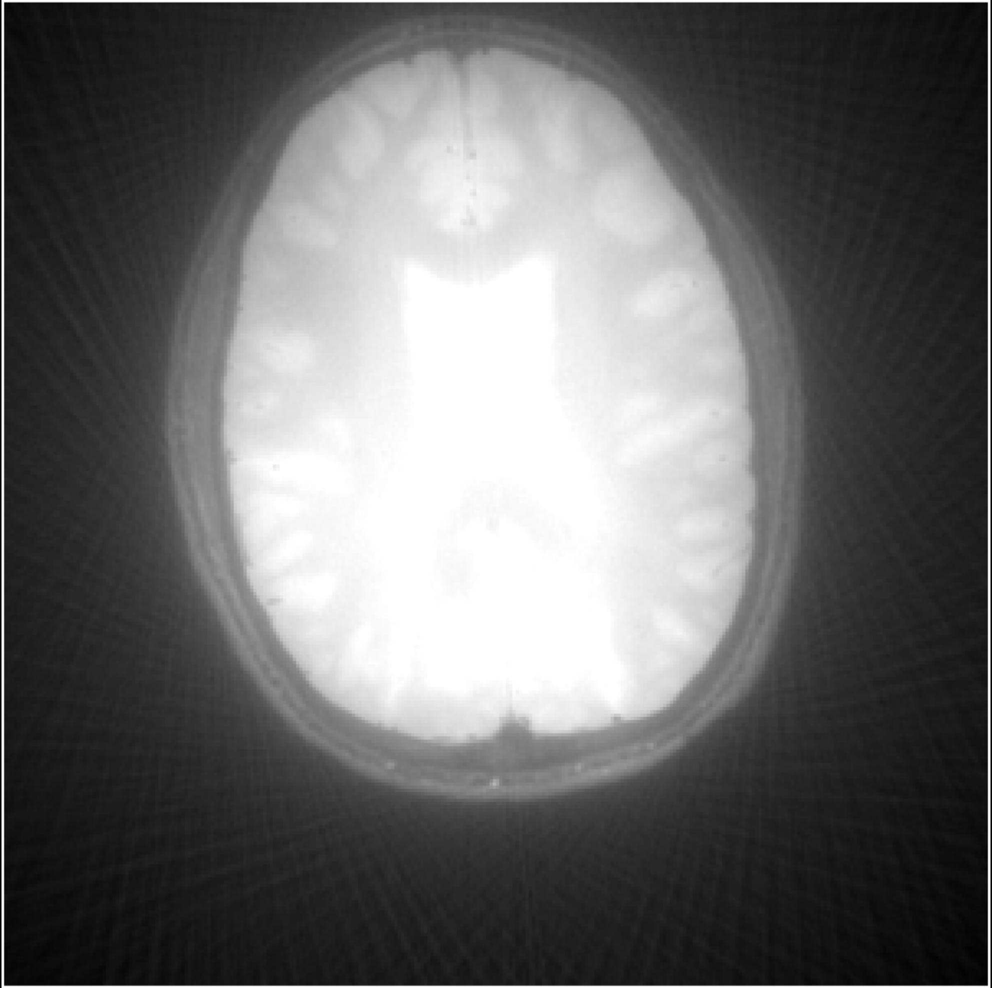
Speaker Name: Hung Do, PhD MSEE

I have the following financial interest or relationship to disclose with regard to the subject matter of this presentation:

Company Name: Canon Medical Systems USA
Type of Relationship: Employee

The *halo* artifact!

Digi Venkat, Wikipedia.org

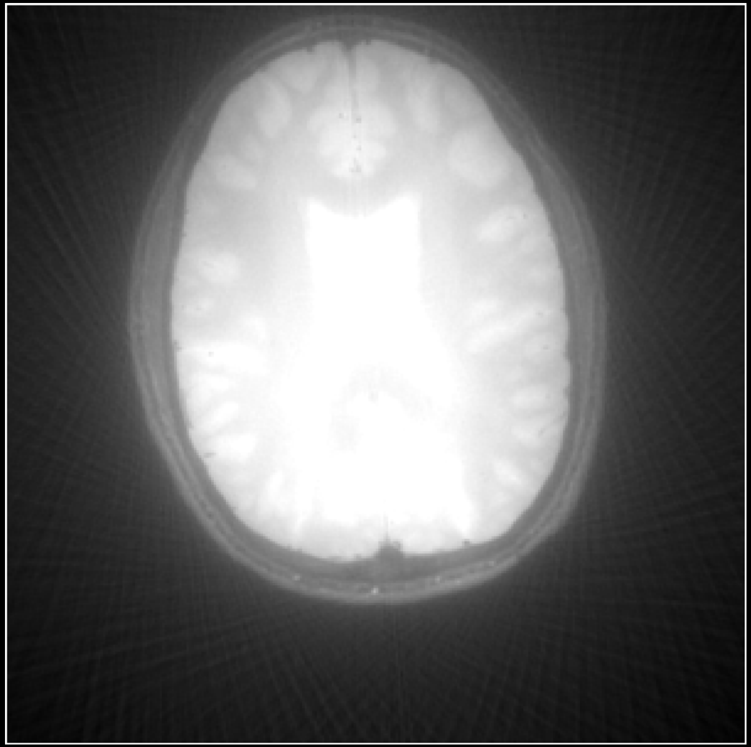


Images from: Hung Do, Canon Medical USA

What causes the *halo* artifact?

A. Corrupted low-frequency k-space
B. Corrupted high-frequency k-space
C. Forget to apply intensity correction
D. Forget to apply density correction

2D single-slice radial imaging with gridding reconstruction



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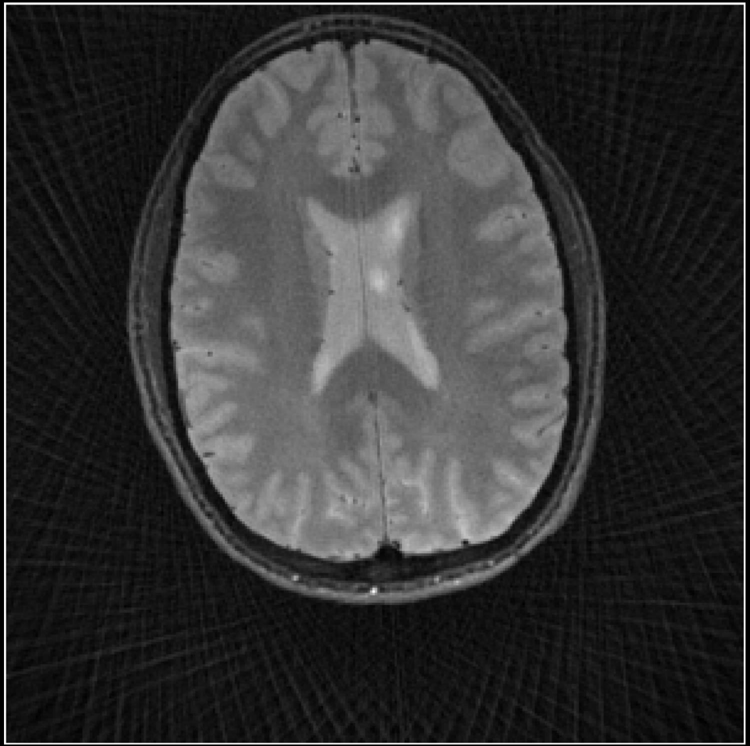
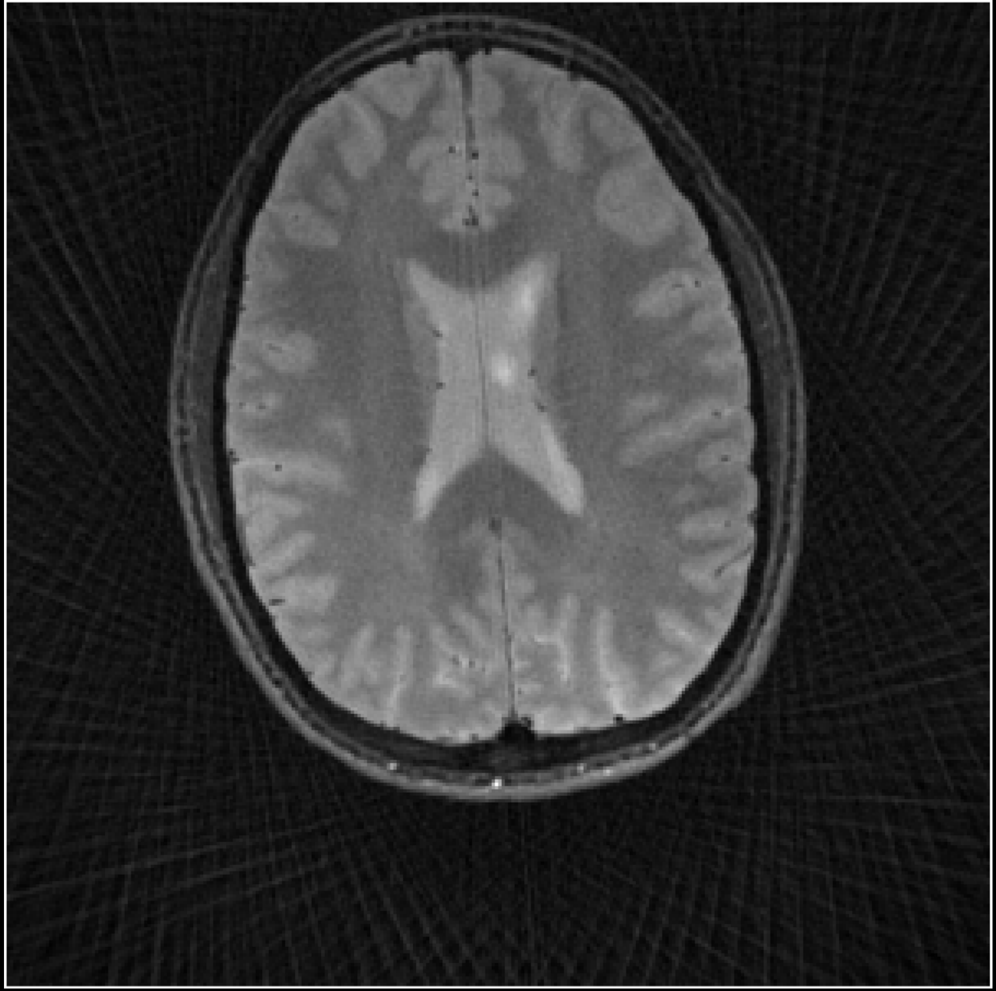


2D single-slice radial imaging with gridding reconstruction

Images from: Hung Do, Canon Medical USA

The streaking artifact!

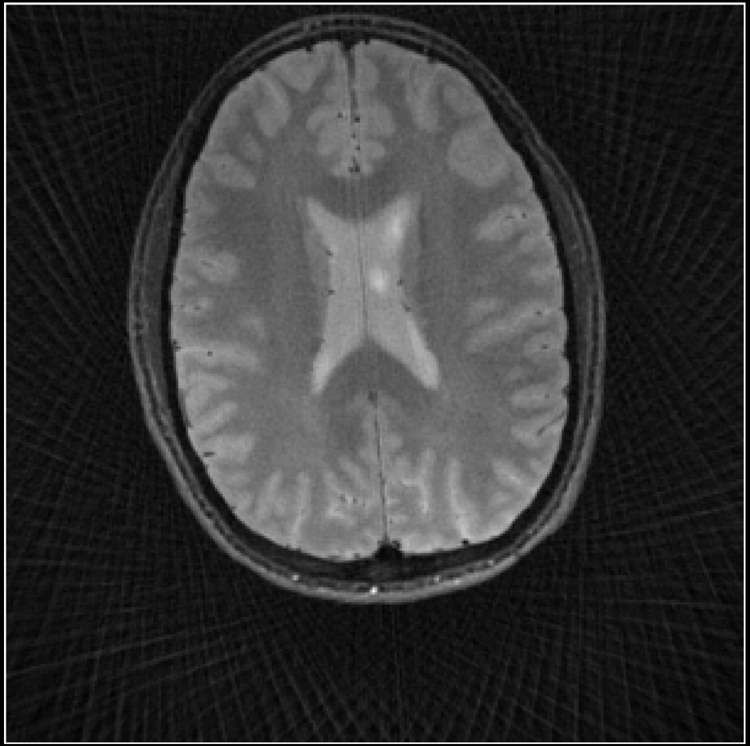


Images from: Hung Do, Canon Medical USA

What causes the *streaking* artifact?

A. Head rotating motions
B. Over-sampled k-space center
C. Under-sampled k-space data
D. Gridding kernel is too narrow

2D single-slice radial imaging with gridding reconstruction



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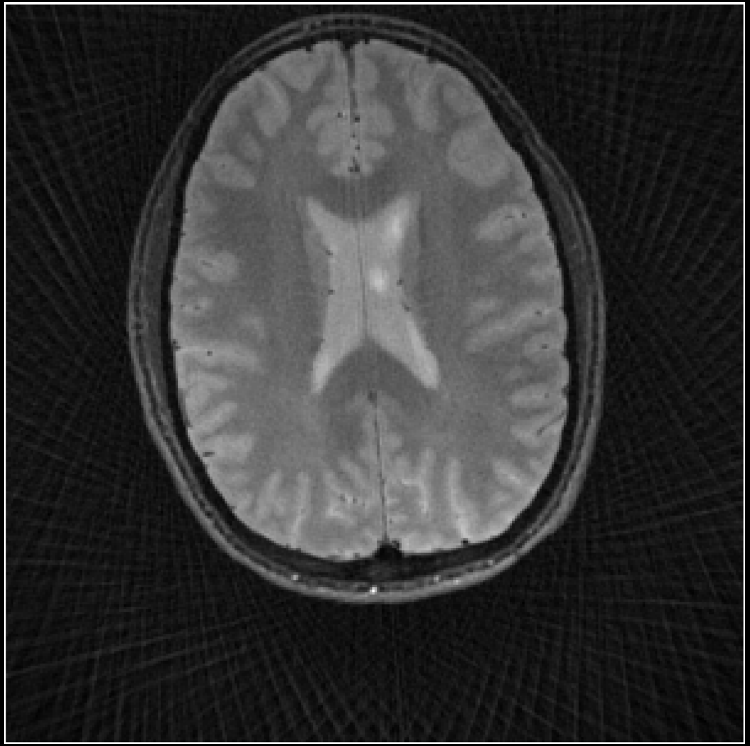
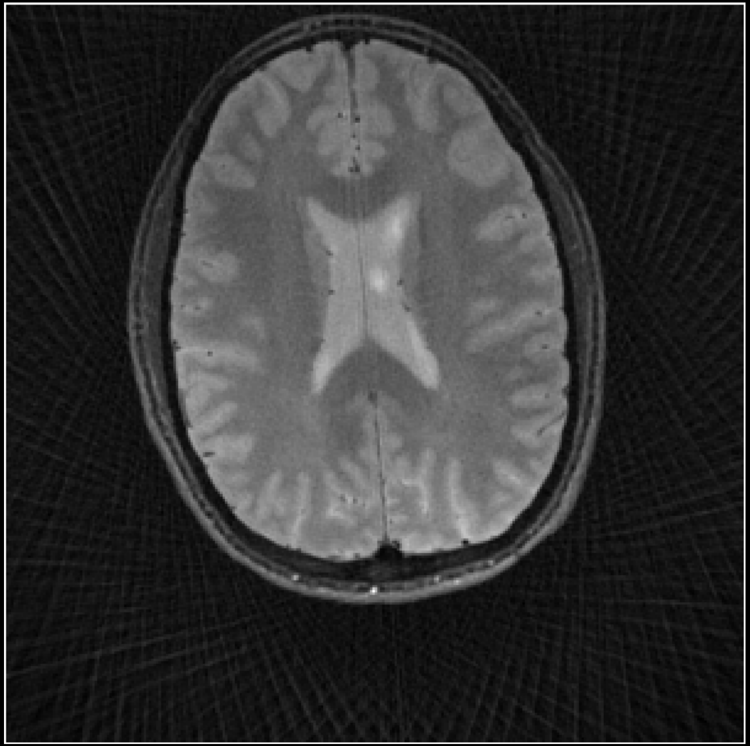


Images from: Hung Do, Canon Medical USA

How can the *streaking* artifact be resolved?

A. Use SENSE (Sensitivity encoding)
B. Use CG-SENSE (Conjugate Gradient SENSE)
C. Use GRAPPA (Generalized Auto calibrating Partial Parallel Acquisition)
D. Use NuFFT (Non-uniform Fast Fourier Transform)



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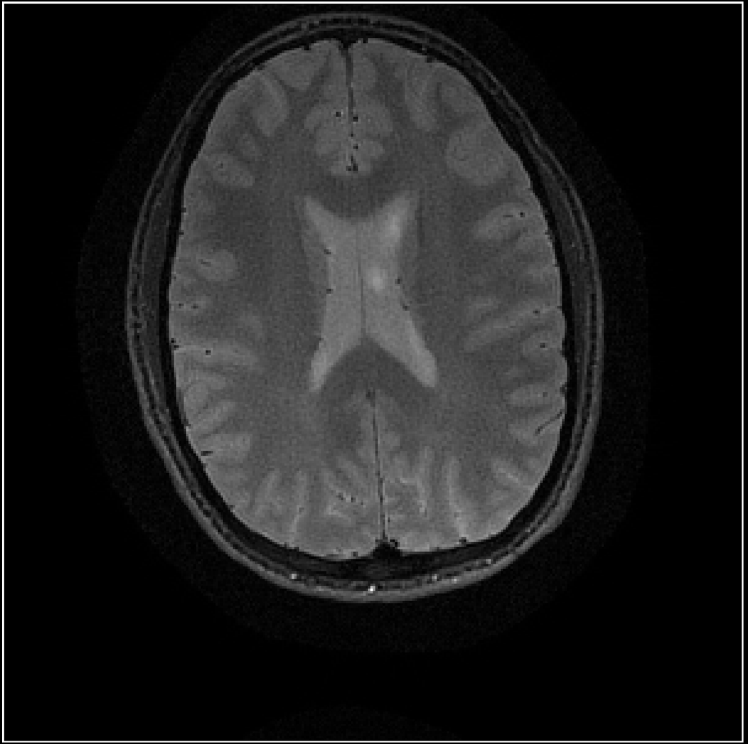


Images from: Hung Do, Canon Medical USA

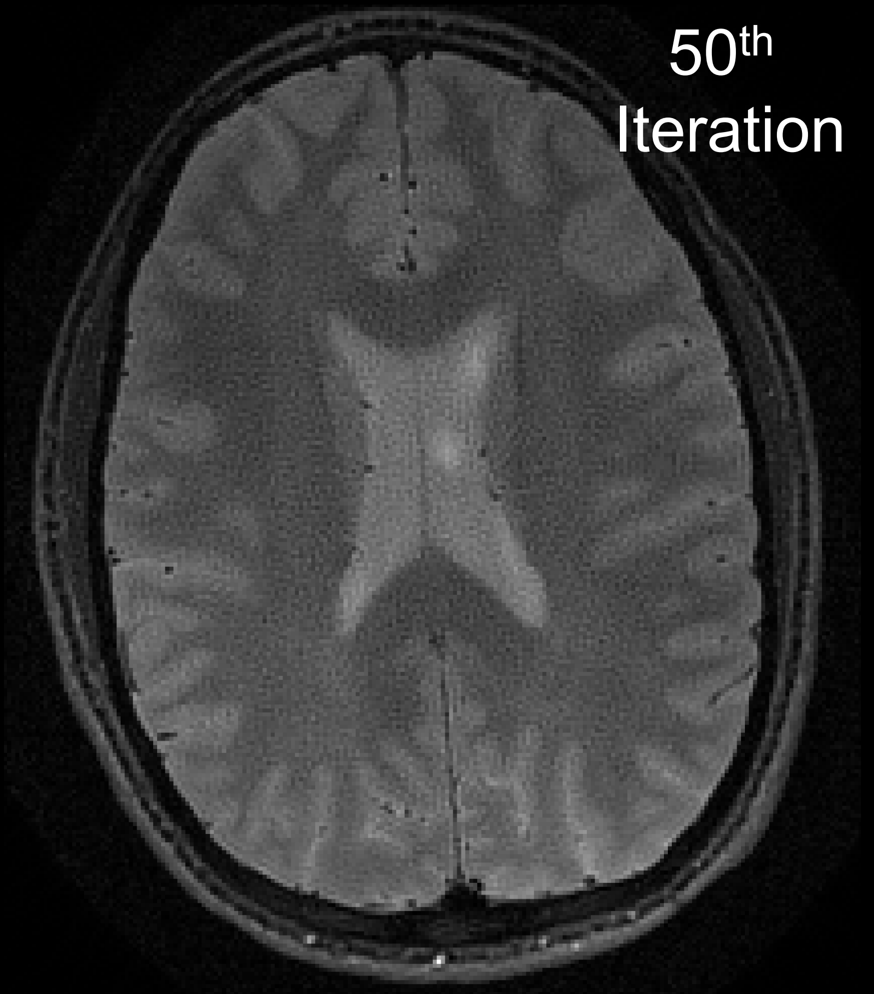
Bonus: Challenging Artifact

The Semi-convergence Behavior of CG-SENSE

What is the Artifact?

A. Remnants of streaking artifacts
B. Over-suppressed background noise
C. Noise amplification
D. Under smoothing

CG-SENSE Recon in action!



50th Iteration

Images from: Hung Do, Canon Medical USA

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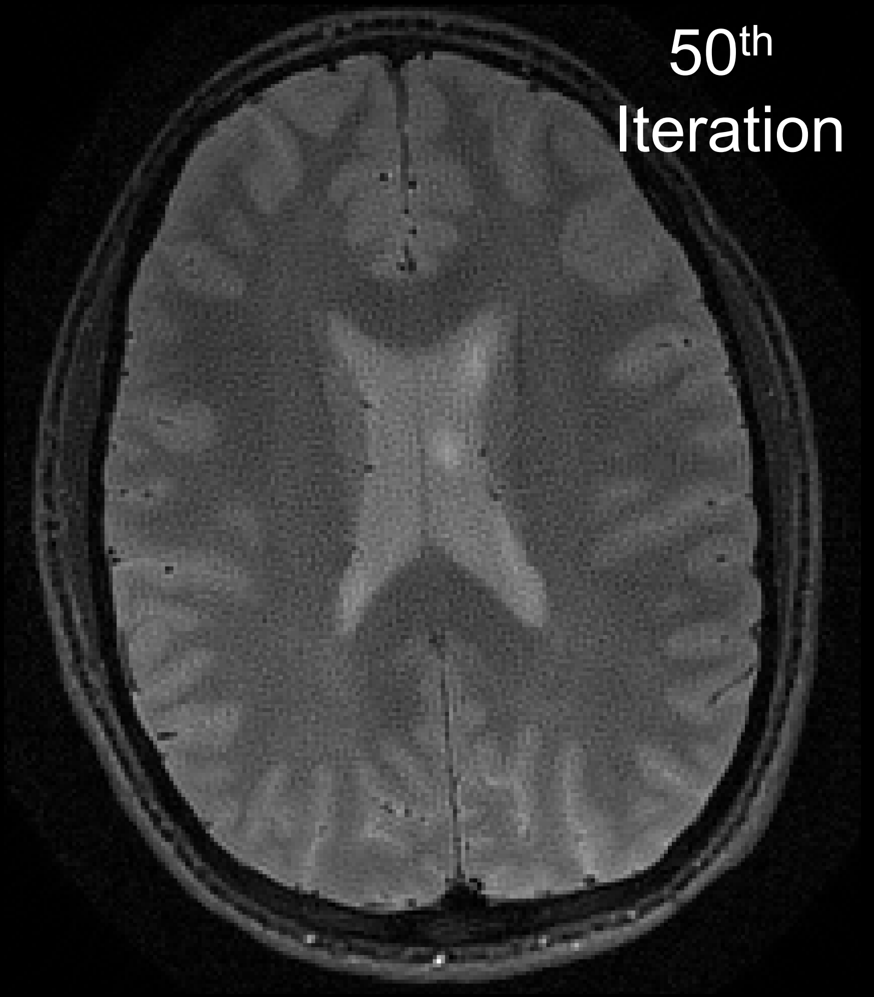
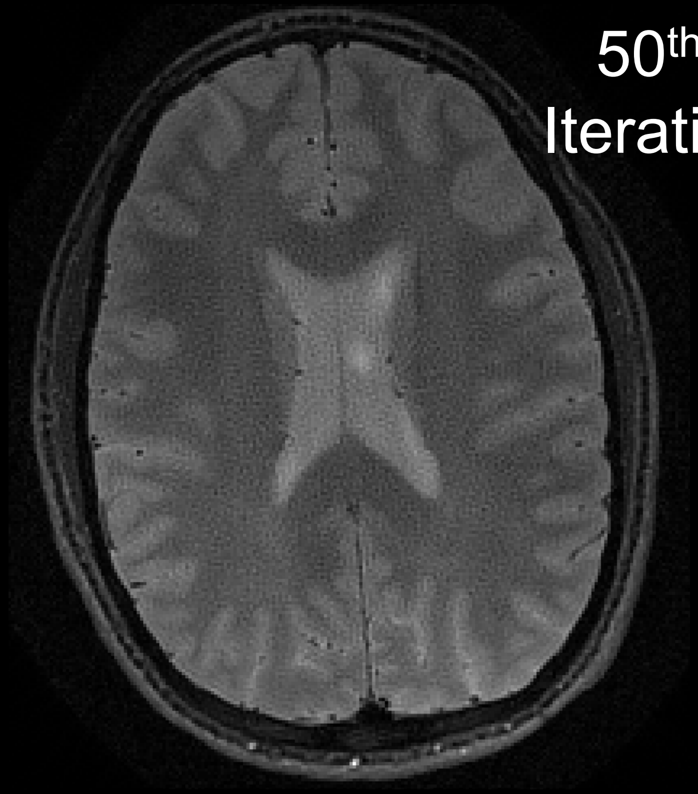
CG-SENSE Recon in action!



Images from: Hung Do, Canon Medical USA

How can the artifact be resolved?

A. Decrease the number of iterations
B. Increase the number of iterations
C. Remove regularization
D. Add noise pre-whitening step to the reconstruction pipeline



CG-SENSE Recon in action!

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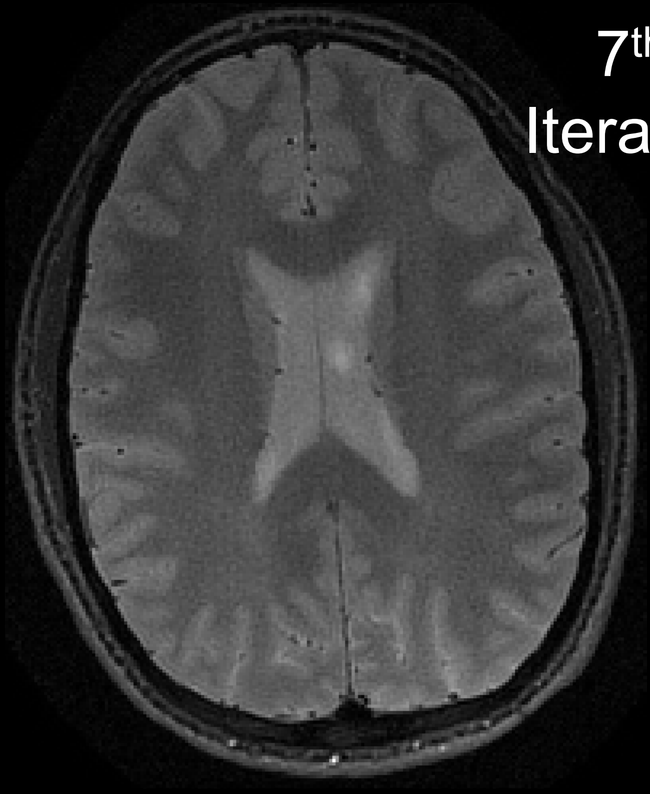

7th Iteration

CG-SENSE Recon in action!

Images from: Hung Do, Canon Medical USA

Artifacts Explanation

Hung Do, PhD MSEE
Canon Medical Systems USA

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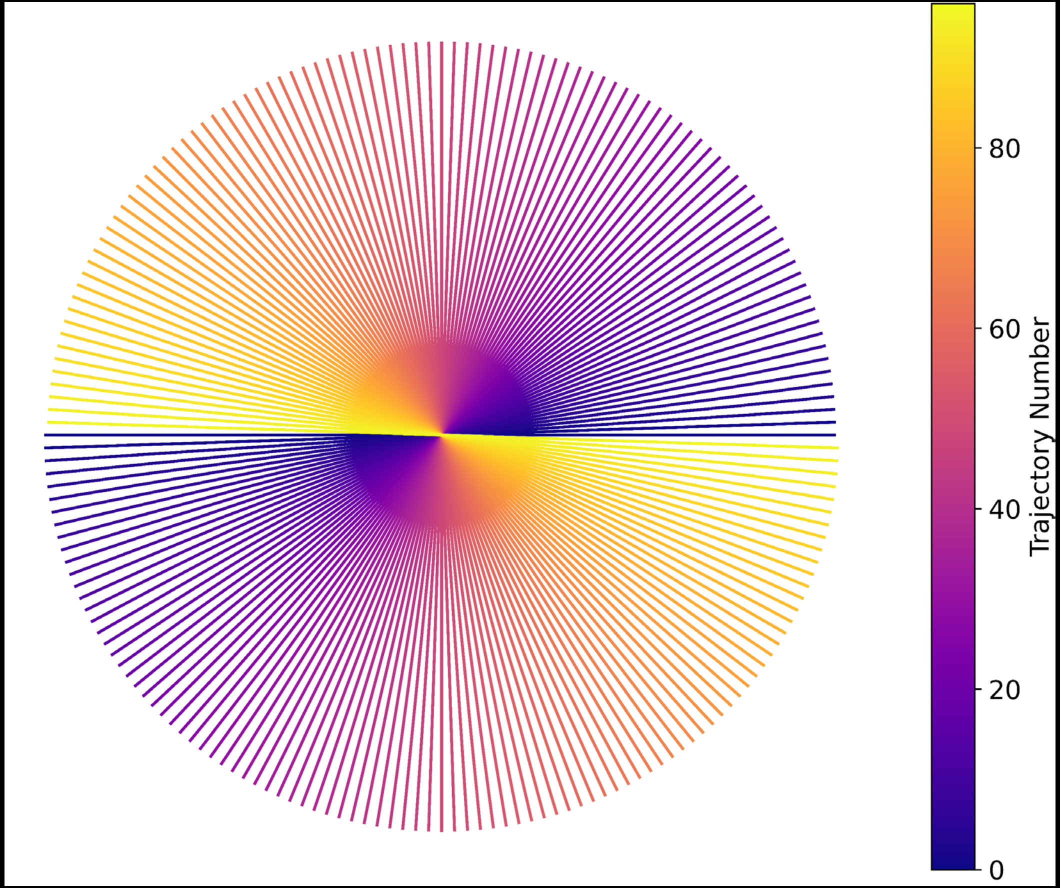
Type of Relationship: **Employee**

Radial k-space trajectory



Non-uniform k-space data:

- Denser low-frequency samples (center)
- Sparser high-frequency samples (periphery)

The non-uniformity must be compensated before reconstruction to avoid the *halo* (low-frequency) artifact.

Gridding reconstruction

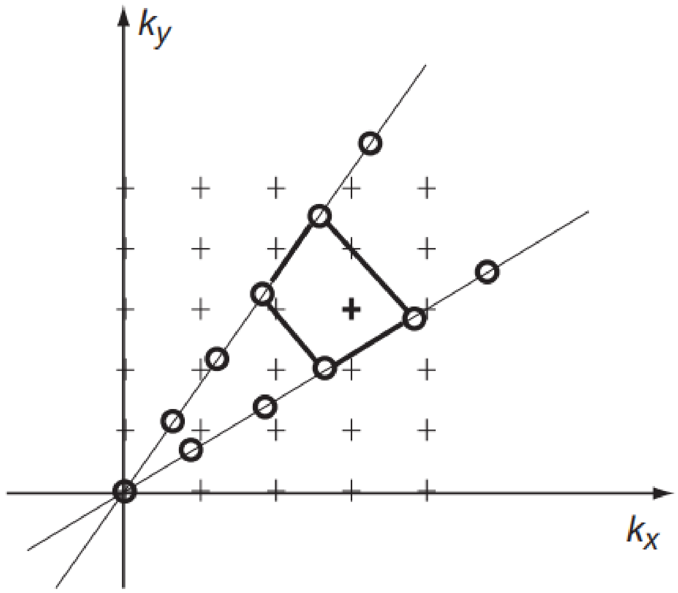


Figure 5.5: Grid-driven interpolation for a projection data set. Data samples lie on diameters in k-space. In this example the surrounding four data samples (o's) are located for each grid point (+'s), and a value at the grid point determined by bilinear interpolation.

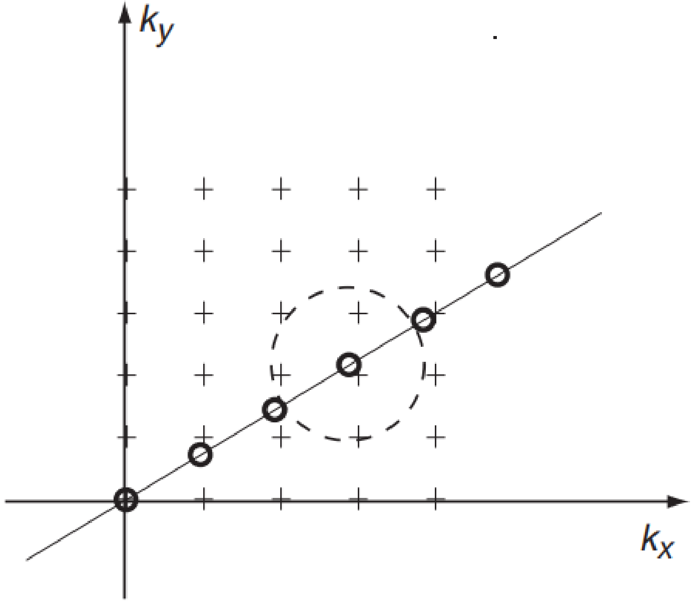


Figure 5.6: Data-driven interpolation for a projection data set. Again, data samples lie on diameters in k-space. Each data point is conceptually considered to be convolved with a small kernel, and the value of that convolution added to the adjacent k-space grid points.

Gridding reconstruction

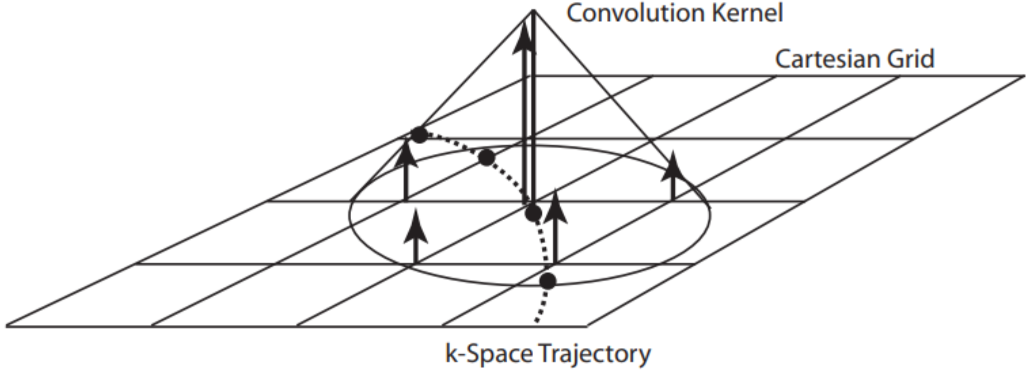
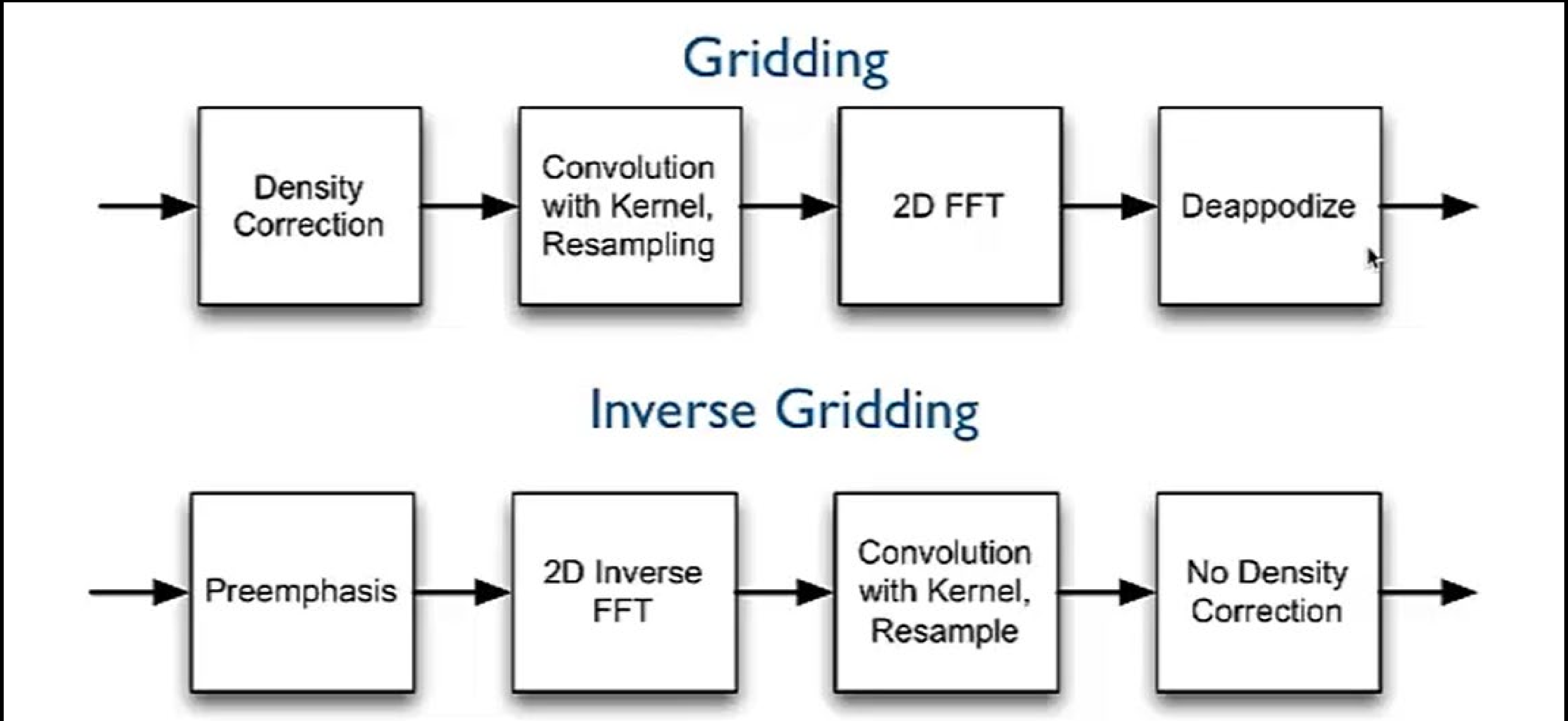


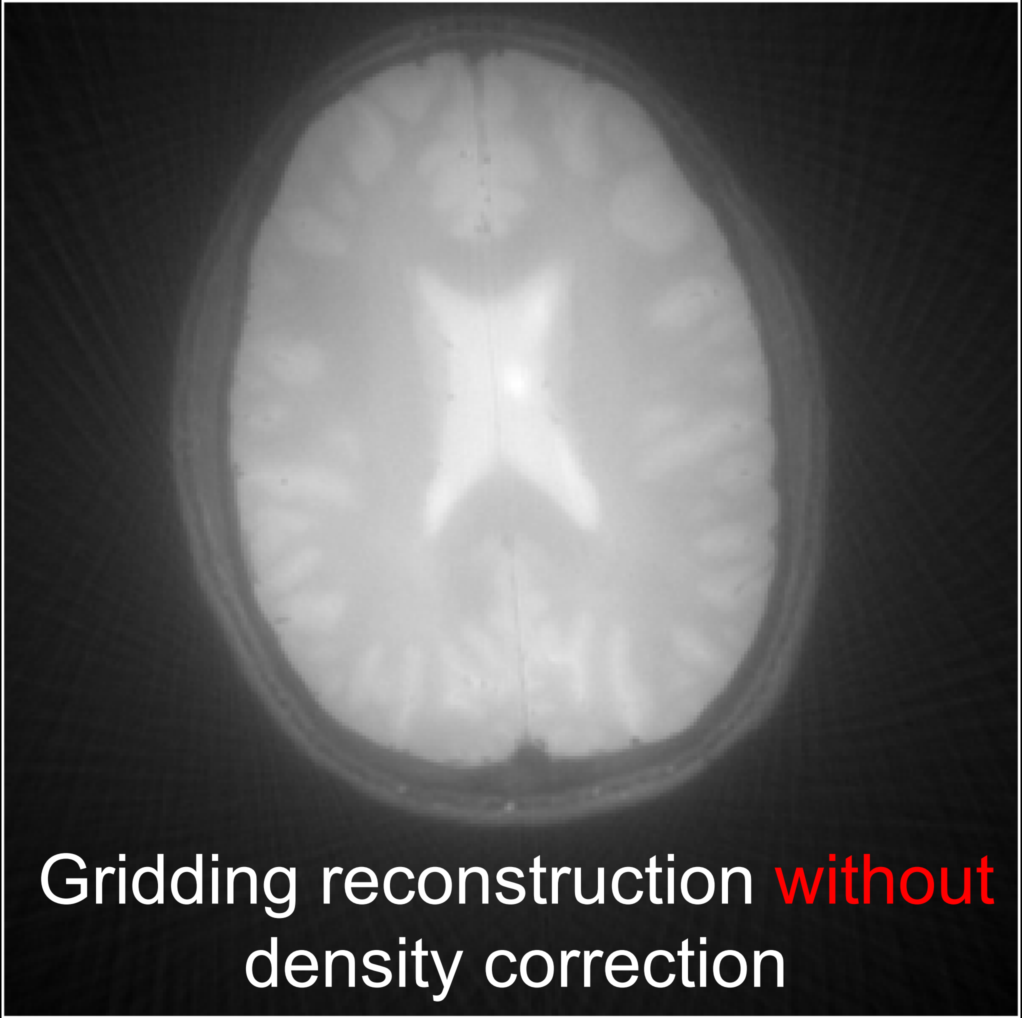
Figure 5.7: Basic gridding idea. The data samples line on some trajectory through k-space (dashed line). Each data point is conceptually convolved with a gridding kernel, and that convolution evaluated at the adjacent grid points.

Gridding & Inverse Gridding

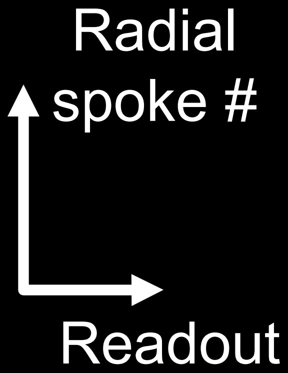


Professor John Pauly's Lecture on "Reconstruction of Non-Cartesian Data", Stanford University

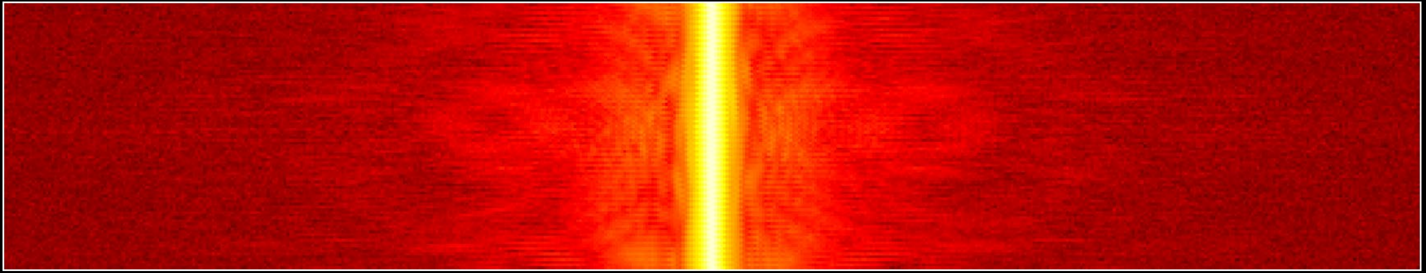
The *halo*
(low-frequency)
artifact



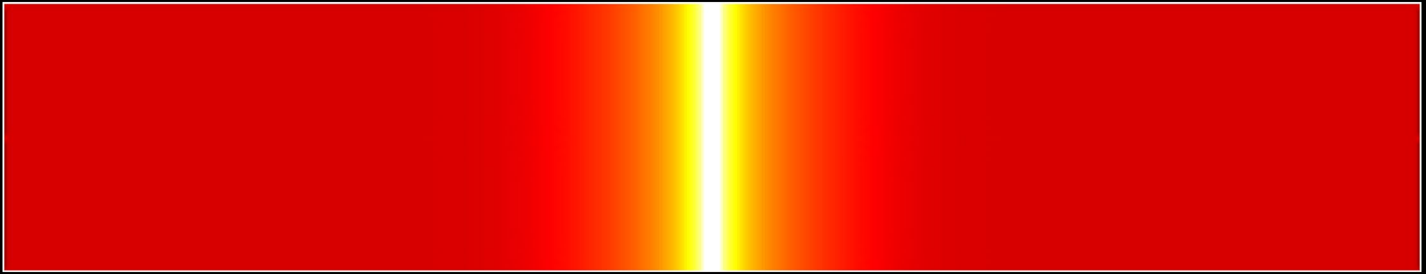
Gridding reconstruction without
density correction



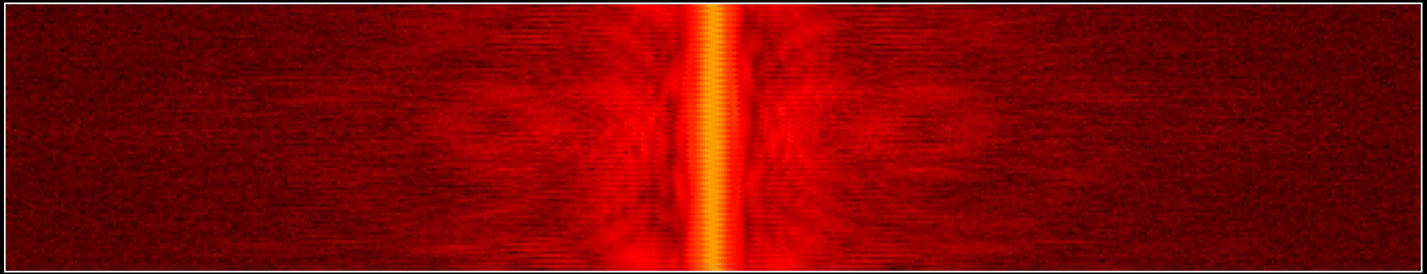
Raw k-space data (**m**)



Density compensation function (**d**)

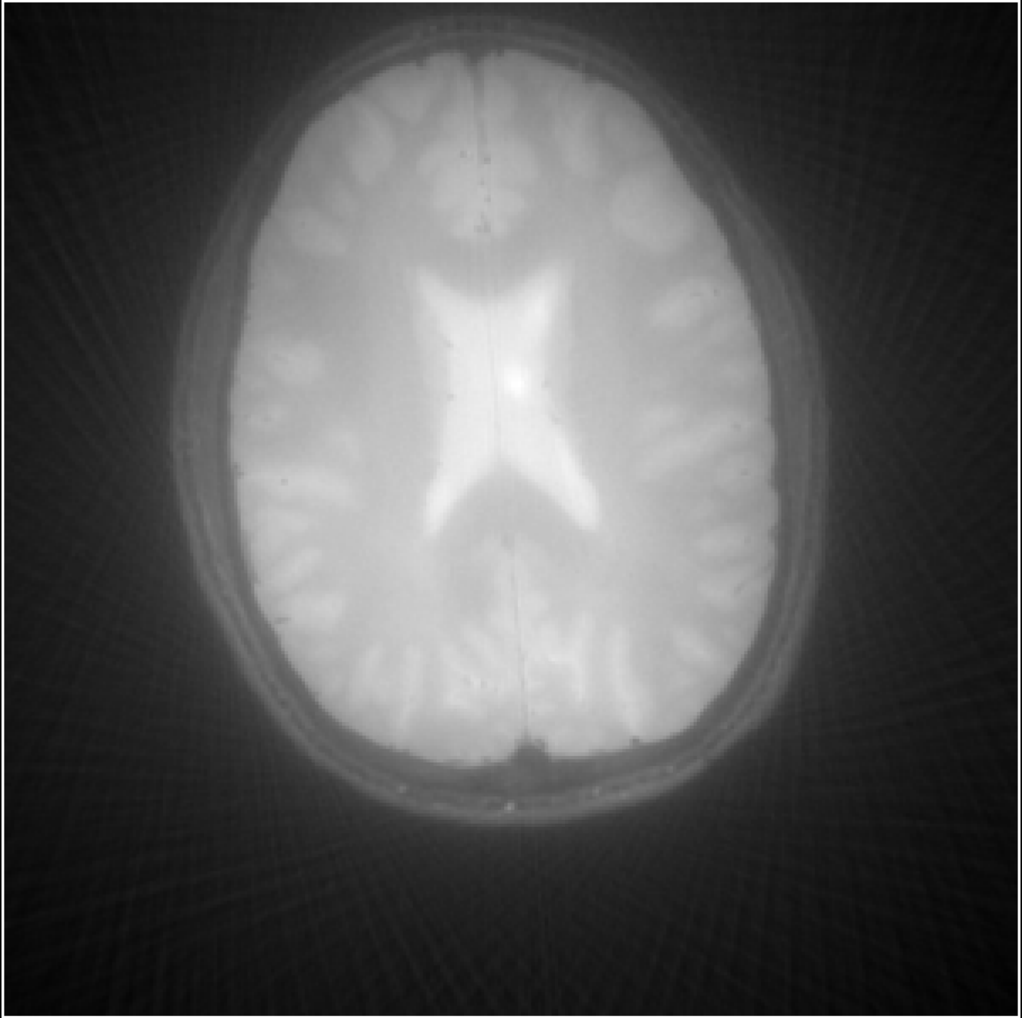


Density corrected data = **m/d**

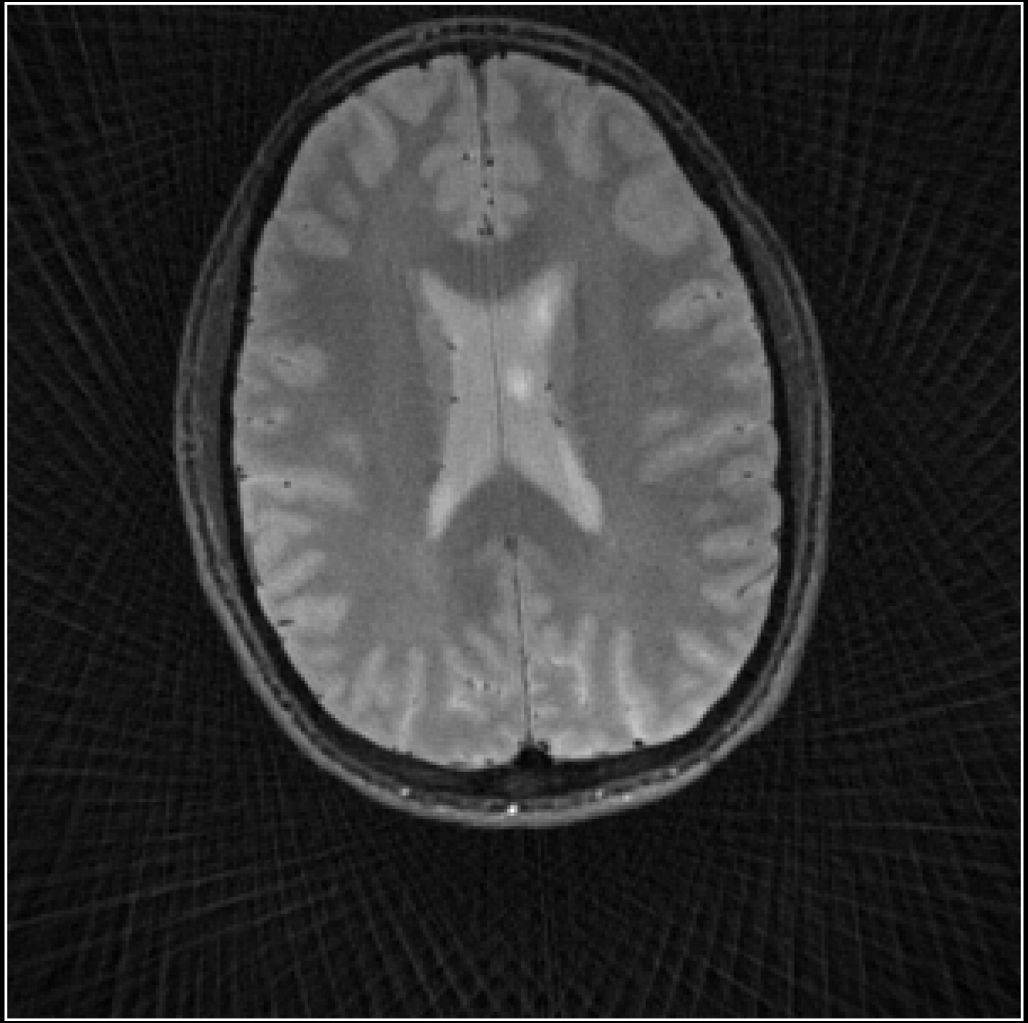


Density
correction

Gridding reconstruction
without
density correction
(halo artifact)

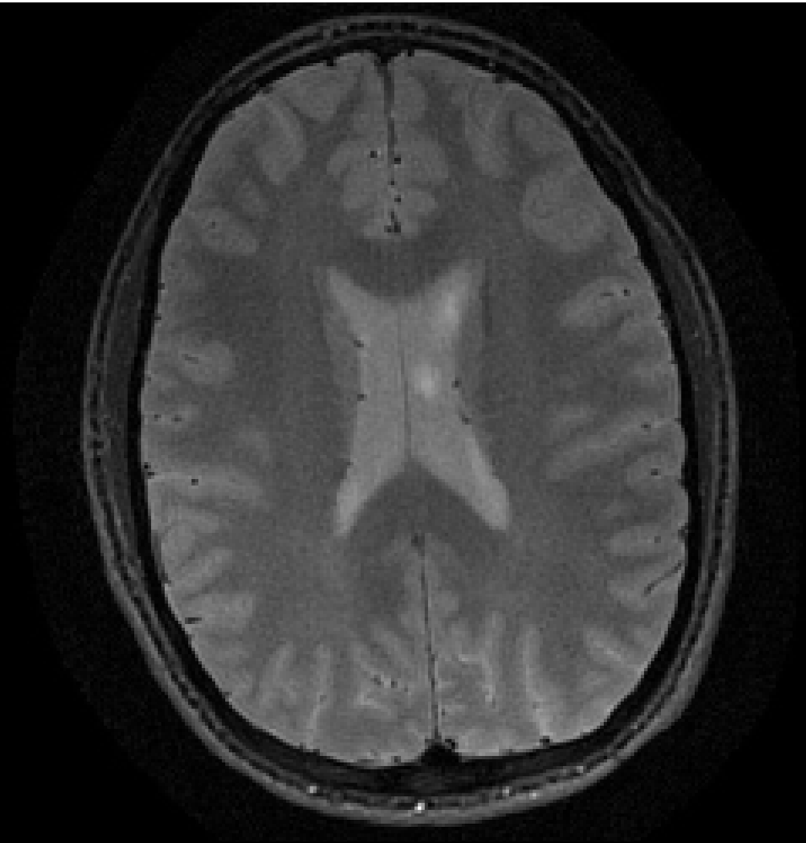


Gridding reconstruction
with
density correction
(streaking and blurring)



SENSE and GRAPPA are for Cartesian k-space while CG-SENSE is for arbitrary k-space trajectories

CG-SENSE reduces streaking artifacts and improves image sharpness for radial MRI



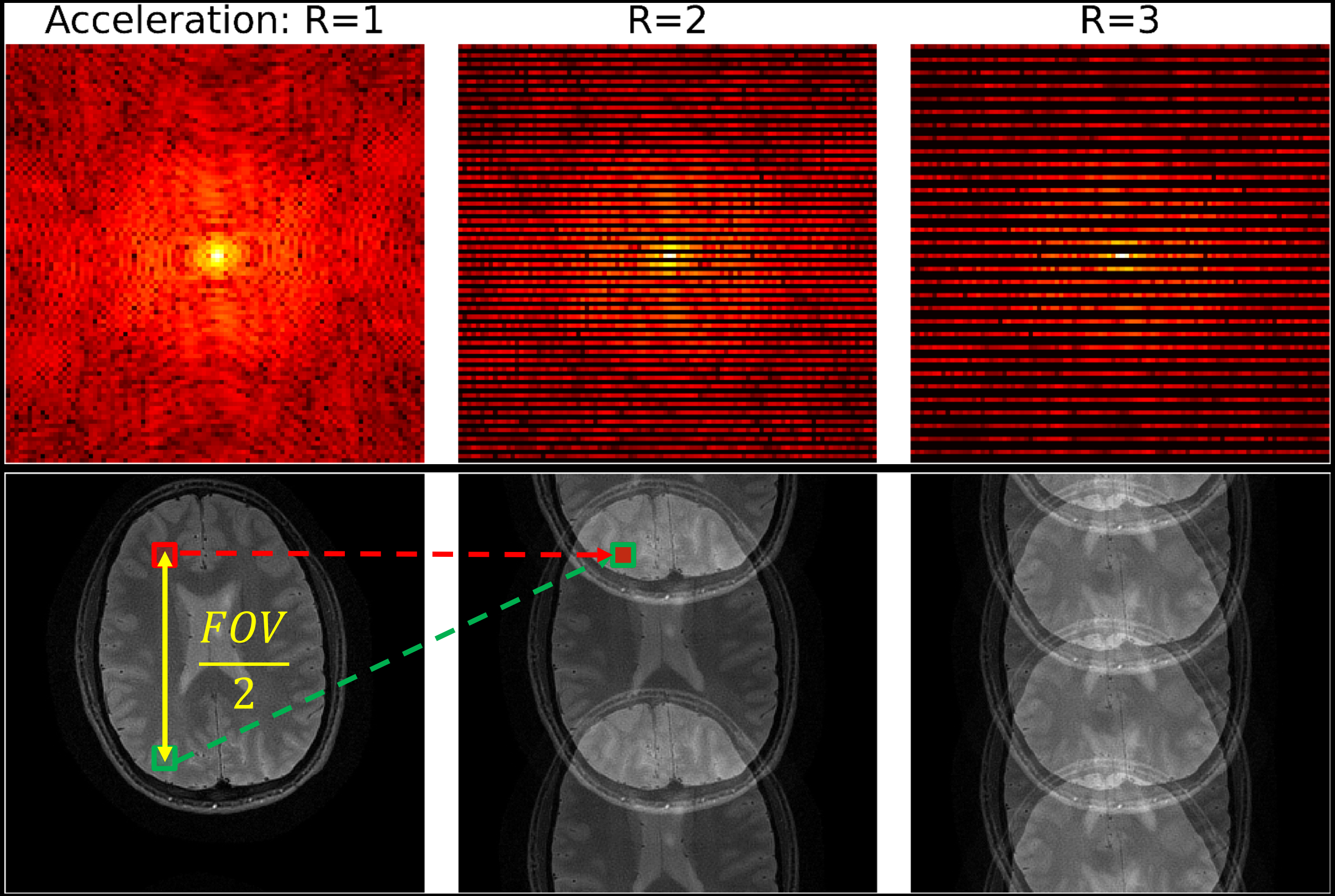
$$\hat{\mathbf{x}} = \mathbf{argmin}\, ||\mathbf{Ex} - \mathbf{m}||_2^2$$

SENSE

Sensitivity Encoding: Parallel Imaging for Cartesian k-space

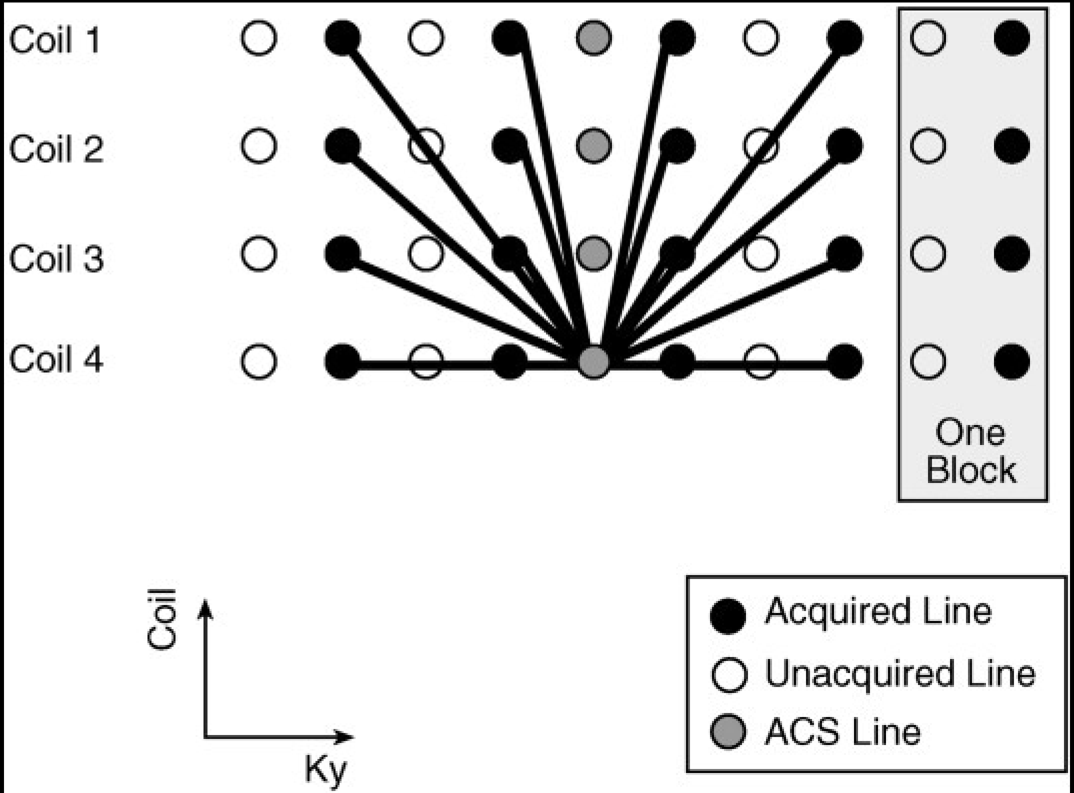
Cartesian k-space

FFT reconstruction



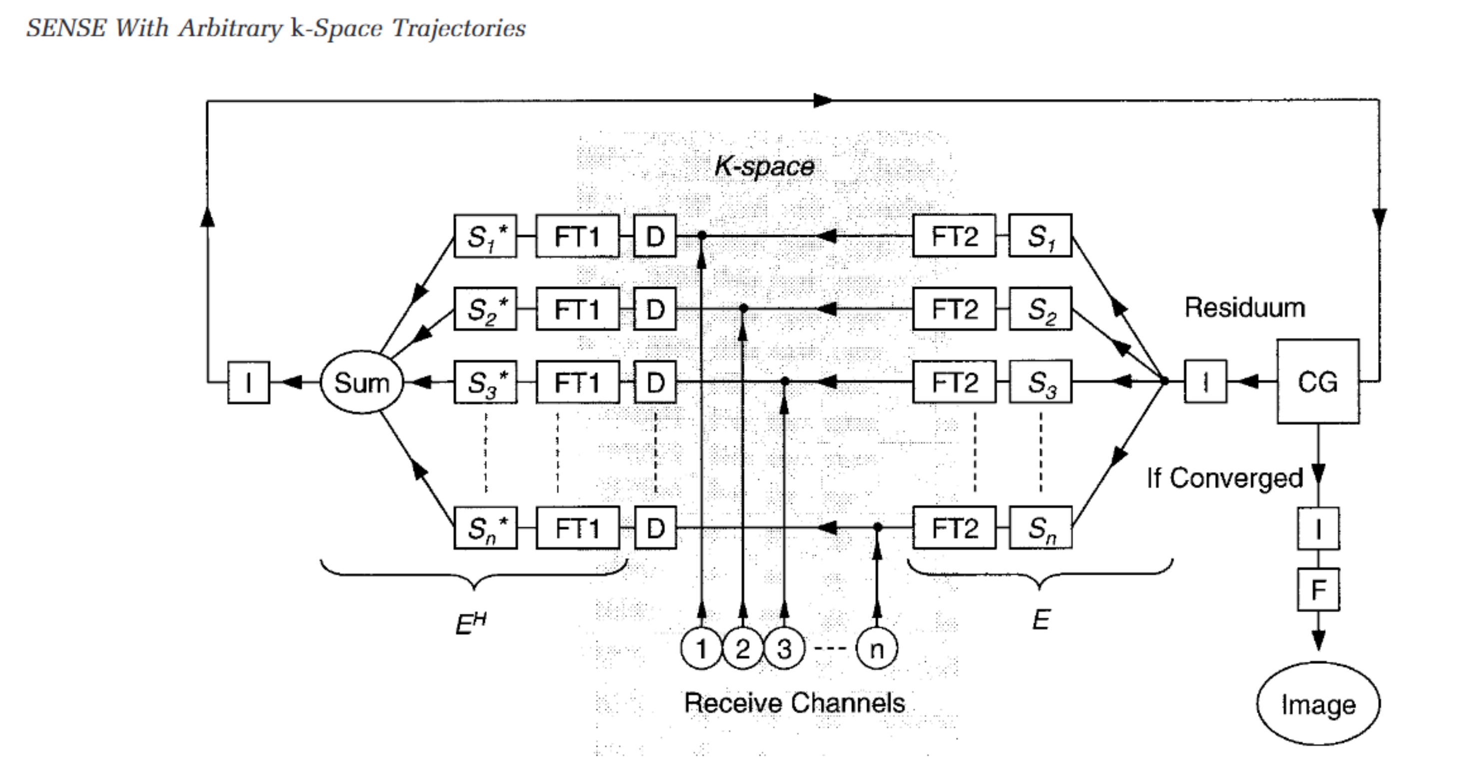
Pruessmann et al., Magn Reson Med. 1999;42:952-962.

Generalized Autocalibrating Partially Parallel Acquisitions (GRAPPA) for Cartesian k-space data



Griswold et al., Magnetic Resonance in Medicine 47.6 (2002): 1202-1210.

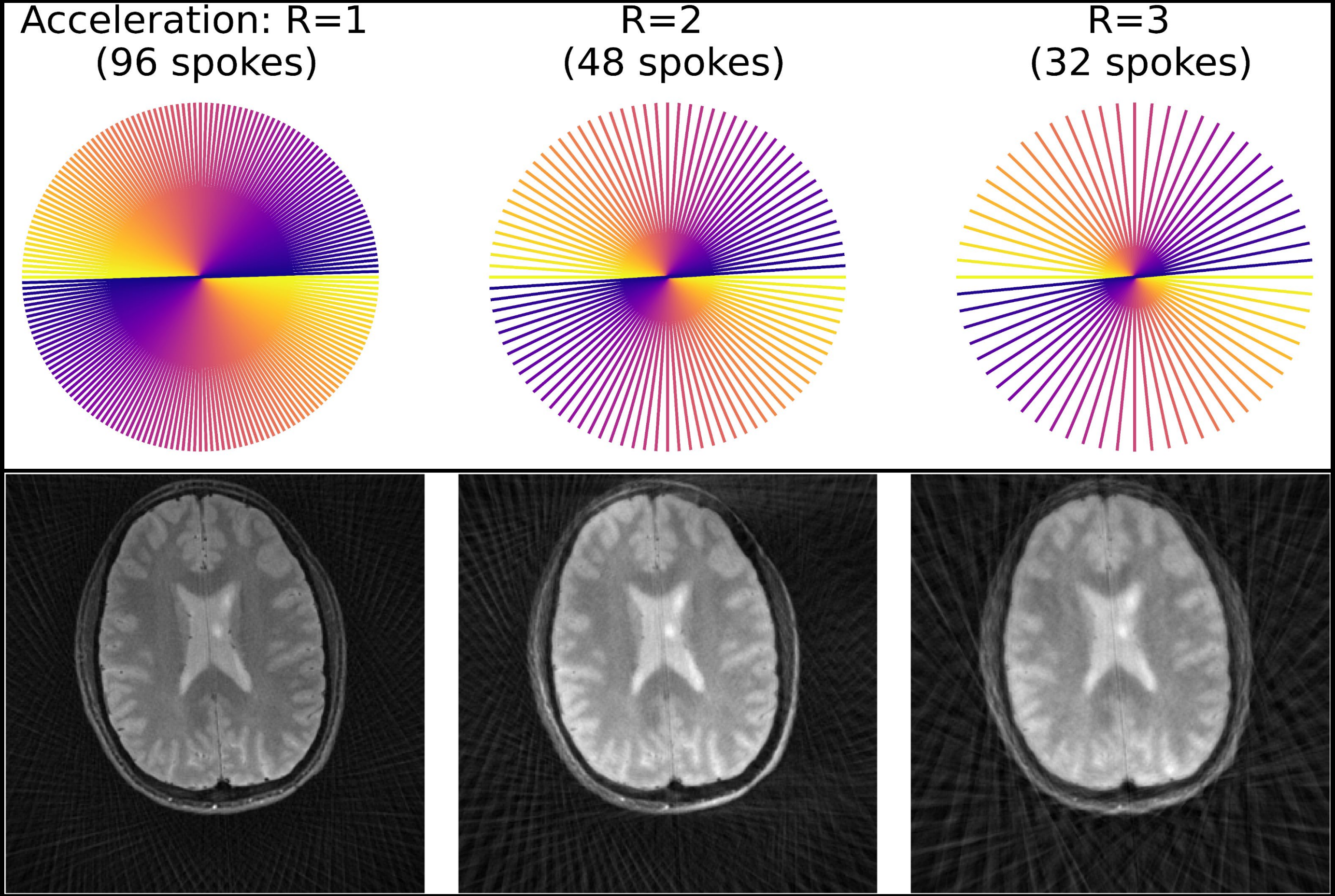
Conjugate Gradient SENSE reconstruction (CG-SENSE) for arbitrary k-space trajectories



Pruessmann et al., Magnetic Resonance in Medicine 46.4 (2001): 638-651.

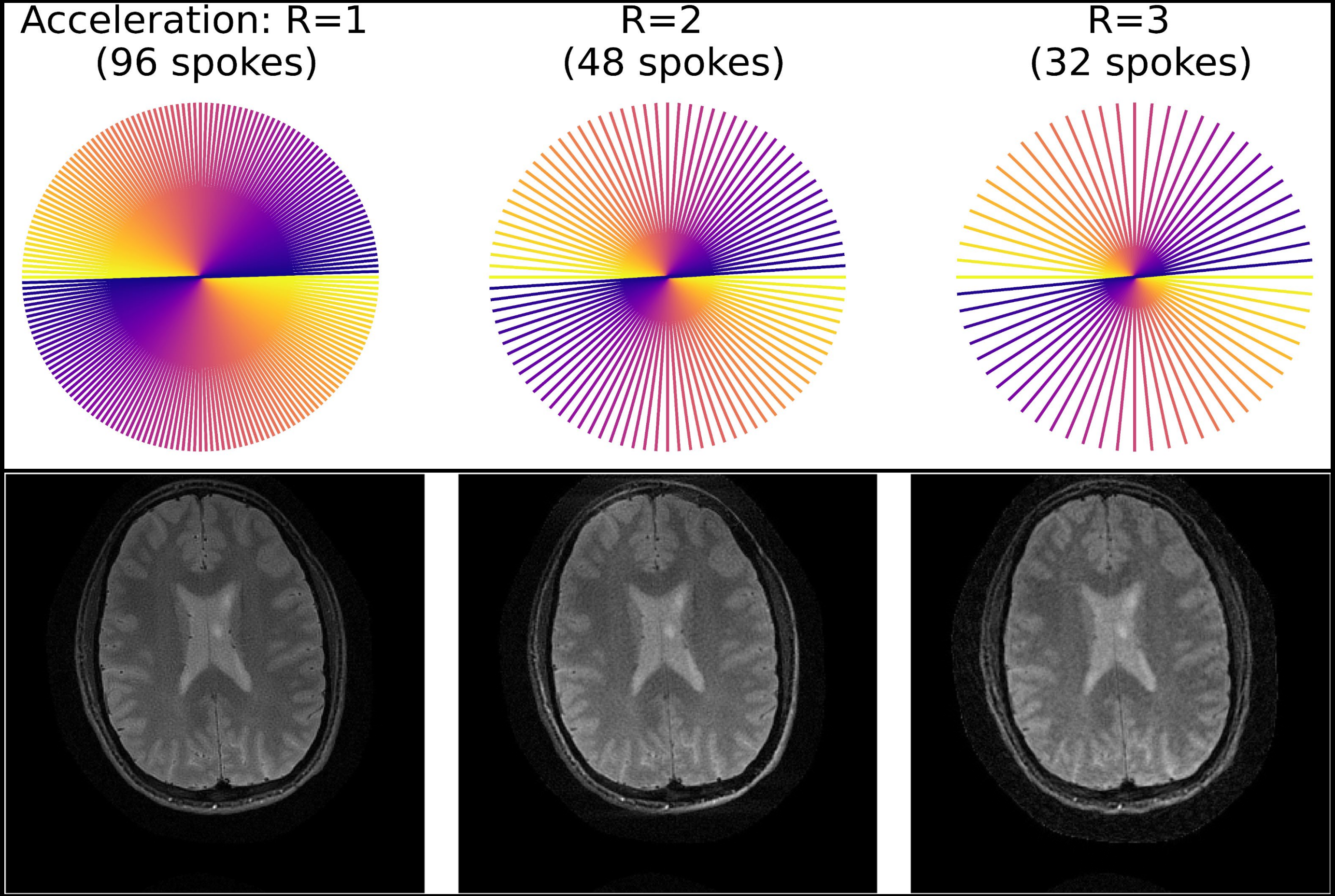
Radial
k-space
trajectory

Gridding
reconstruction
(streaking &
blurring)



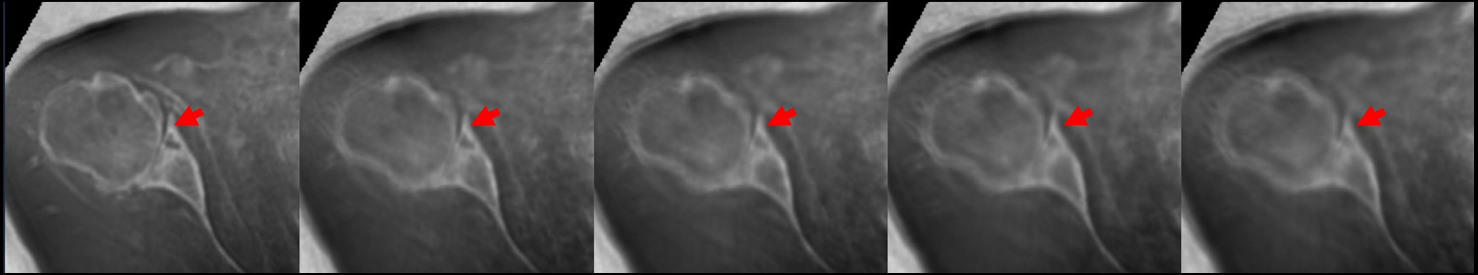
Radial k-space trajectory

CG-SENSE reconstruction

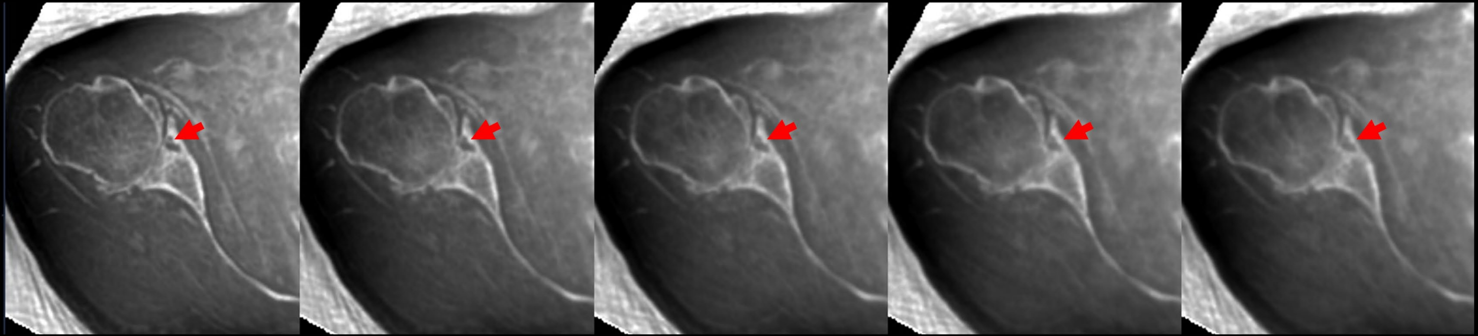


Utilizing CG-SENSE in clinical setting

GRID: 5.8min | 2.9min | 1.9min | 1.4min | 1.2min

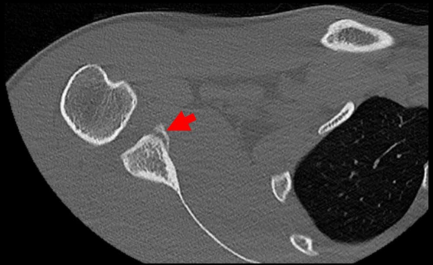


DLR: 5.8min | 2.9min | 1.9min | 1.4min | 1.2min



Bony Bankart Lesion (arrow)

CT



Hung Do *et al.*, ISMRM 2025, Abstract #0156

4-echo UTE, 0.8mm^3 with CG-SENSE & Deep Learning Denoising Recon (DLR) vs. Gridding

Want to know more?

https://global.medical.canon/products/magnetic-resonance/good-to-know#ultrashort

Want to know more?

Canon

Good to Know

MRI

Advanced intelligent Clear-IQ Engine (AiCE)

Hung P. Do, PhD

WHAT? A deep learning based reconstruction method for MRI that intelligently removes noise while maintaining feature integrity.

WHY? To increase SNR of the reconstructed images. This increased SNR could be translated to increased resolution and/or shortened scan time. This could also enable high field-like image quality without high-field challenges (e.g. higher cost, B_0 & B_1 inhomogeneity, etc.).

WHEN? Applicable to all anatomies and available at both 1.5T and 3T, for both wide and narrow bore system.

https://global.medical.canon/products/magnetic-resonance/good-to-know#advanced

Want to know more?

Canon

Good to Know

MRI

Conjugate Gradient Reconstruction
CG Recon

Chia-Ying Liu, PhD

WHAT? CG Recon is a reconstruction technique that allows parallel imaging to be applied for non-Cartesian sampling acquisitions

WHY? The aliasing that results from undersampled data is considerably more complex in non-Cartesian acquisitions and prevents straightforward unfolding by conventional parallel imaging methods

WHEN? CG Recon enables parallel imaging for radial ultra-short echo time (UTE) imaging

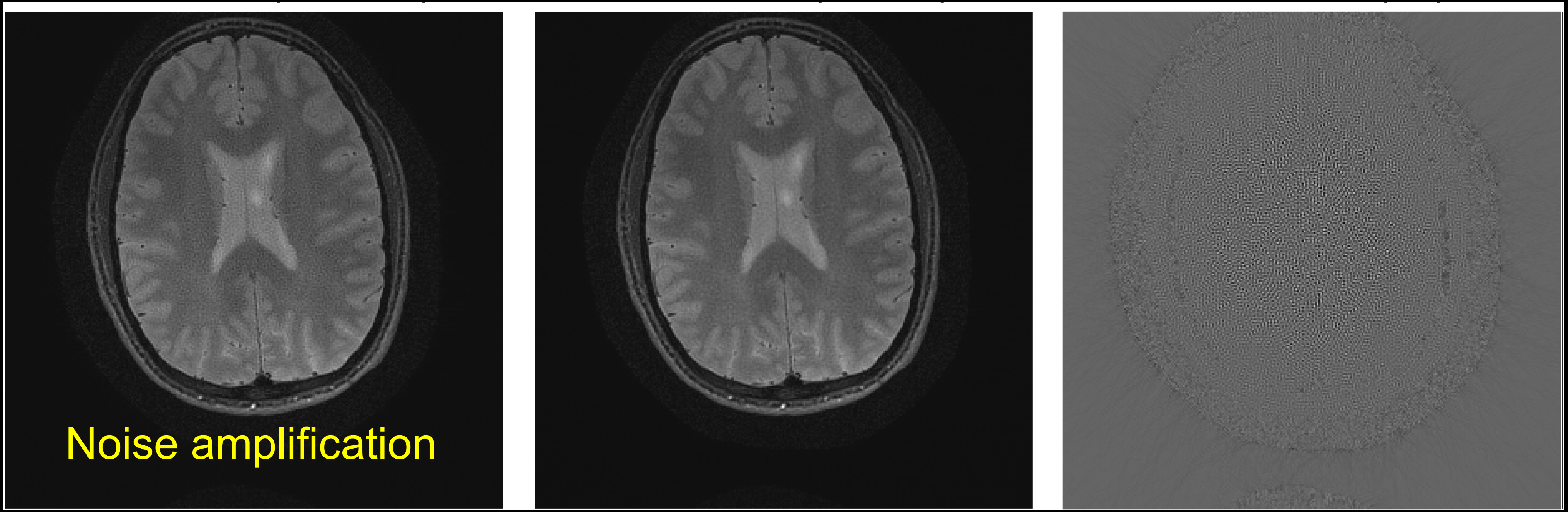
https://global.medical.canon/products/magnetic-resonance/good-to-know#cgrecon

Bonus: Challenging Artifact

The Semi-convergence Behavior of CG-SENSE

Noise amplification with a high number of iterations

50th iteration | 7th iteration | difference (10x)

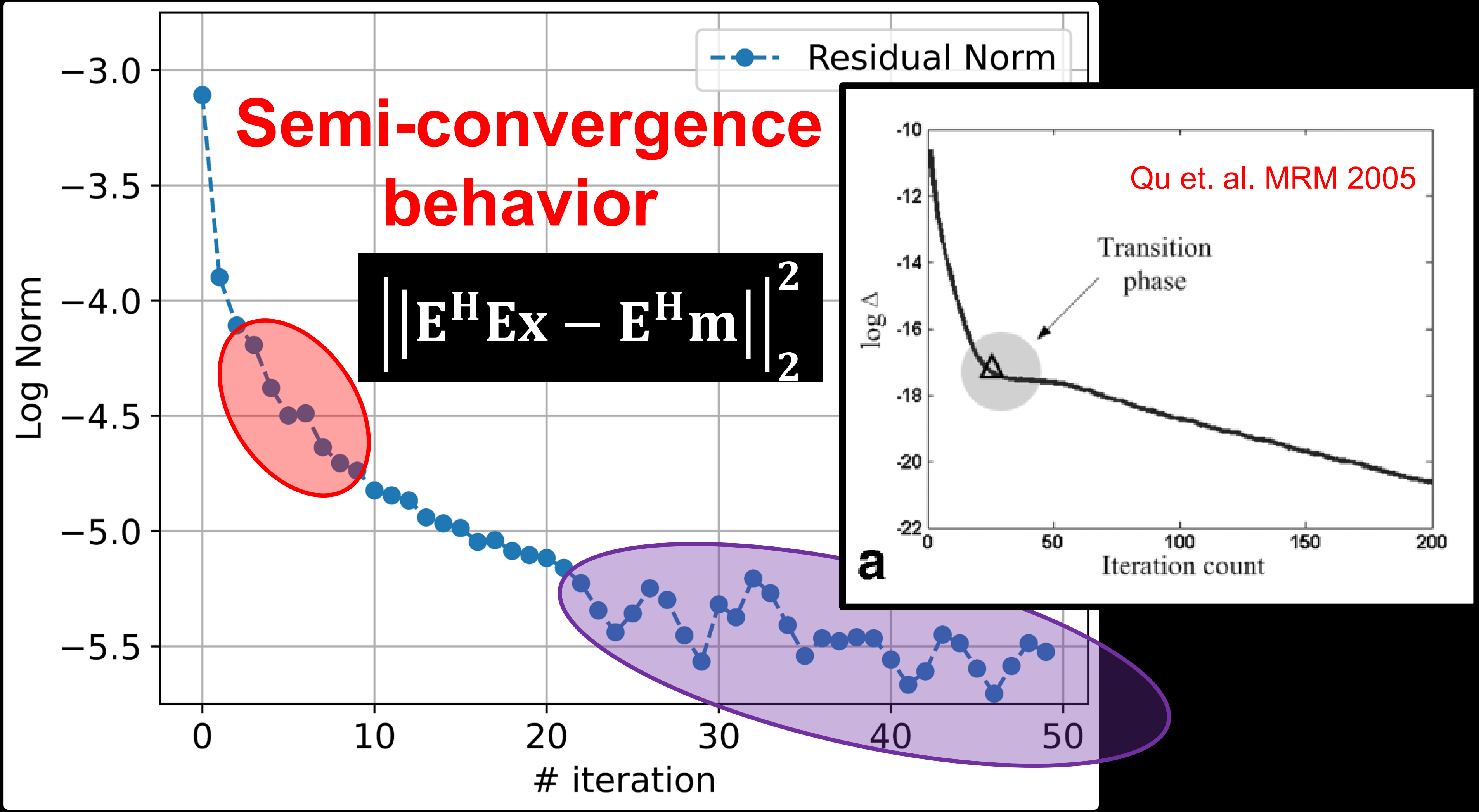


CG-SENSE's Semi-convergence

- *"Semi-convergence is characterized by initial convergence toward the optimal solution but later divergence"* (Qu *et al*., MRM 2005).
- The under-sampled k-space data causes the inverse problem ill-posed, which in turns leads to the semi-convergence behavior.
- The semi-convergence is a feature of the Conjugate Gradient or Gradient Descent algorithm when a problem is ill-posed.

Troubleshooting

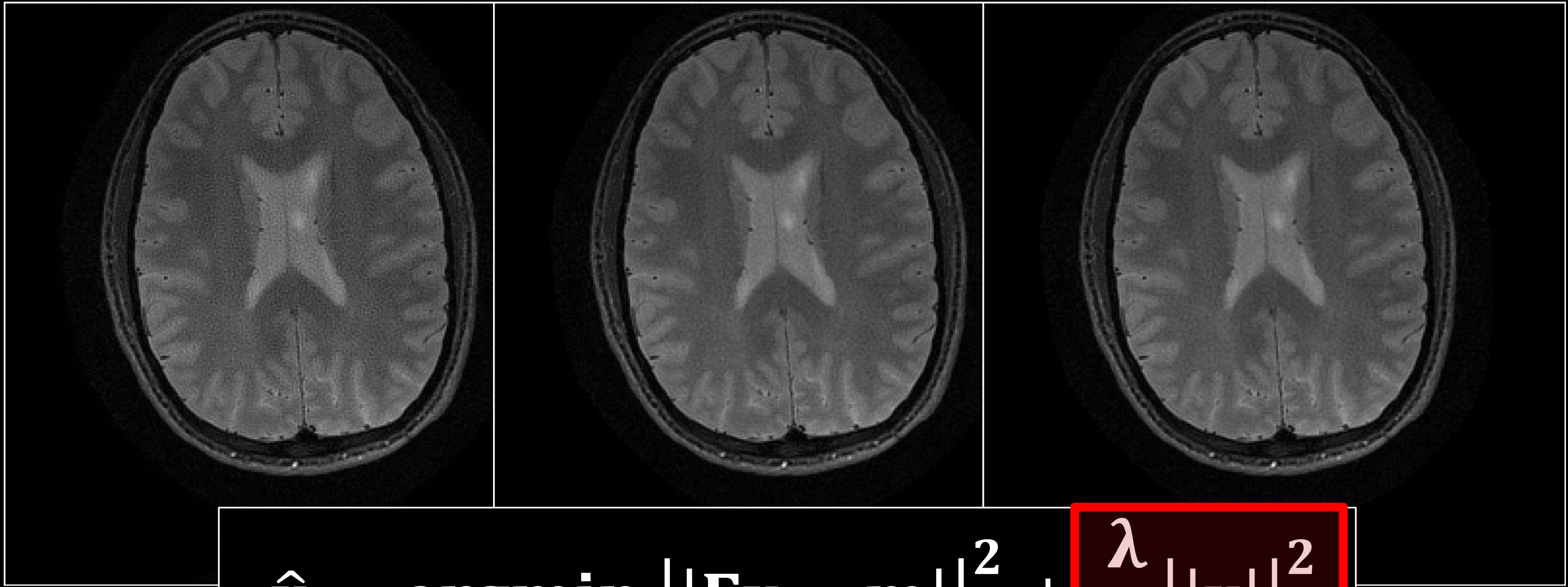
1. My first response, when I saw the grainy (noise-like) artifacts, was to increase the number of iterations, hoping that the reconstructed image would come closer toward the optimal solution (un-aware of the *semi-convergence* behavior). However, the noise amplification got worse.
2. Second, I wondered if there is a bug in the CG implementation. I decided to reimplement different variants of the CG algorithm, but the artifacts remained.
3. Third, I implemented gradient descent algorithm, but the artifacts persisted.
4. Fourth, I inspected the intermediate images and plotted residual norm vs. number of iterations. Residual norm behaved funny at high iteration count, but I wasn't sure why.
5. Finally, I found the Qu *et al.*, MRM2005 and learned about the *semi-convergence* phenomenon.
6. The solution is to use early stopping or to implement CG with regularizations.


Residual Norm
Semi-convergence behavior
$||E^H Ex - E^H m||_2^2$
Log Norm
iteration
Qu et. al. MRM 2005
Transition phase
log Δ
Iteration count
a

Noise amplification

Tikhonov regularization

Early stopping



$$\hat{\mathbf{x}} = \mathbf{argmin}\, \|\mathbf{Ex} - \mathbf{m}\|_2^2 + \boxed{\frac{\boldsymbol{\lambda}}{\mathbf{2}} \|\mathbf{x}\|_2^2}$$

Other Regularizations

Early stopping and Tikhonov regularization mitigate the noise amplification, other advanced regularizations also work:

- L1-regularization in the sparsifying transform domain as in Compressed Sensing reconstruction,
- Data-driven regularization as in Deep Neural Network-based reconstruction, etc.

$$\hat{\mathbf{x}} = \mathbf{argmin}\,||\mathbf{Ex} - \mathbf{m}||_2^2 + \mathbf{R(x)}$$

m: measured k-space
E: encoding matrix
x: intermediate reconstructed image
R(x): regularization function

References

1. Qu, Peng, et al. "Convergence behavior of iterative SENSE reconstruction with non-Cartesian trajectories.“Magnetic Resonance in Medicine 54.4 (2005): 1040-1045.
2. Pruessmann, Klaas P., et al. "Advances in sensitivity encoding with arbitrary k-space trajectories.“ Magnetic Resonance in Medicine 46.4 (2001): 638-651.
3. Maier, Oliver, et al. "CG-SENSE revisited: Results from the first ISMRM reproducibility challenge." Magnetic Resonance in Medicine 85.4 (2021): 1821-1839.
4. Griswold, Mark A., et al. "Generalized autocalibrating partially parallel acquisitions (GRAPPA)." Magnetic Resonance in Medicine 47.6 (2002): 1202-1210.
5. Lustig, Michael, David Donoho, and John M. Pauly. "Sparse MRI: The application of compressed sensing for rapid MR imaging." Magnetic Resonance in Medicine 58.6 (2007): 1182-1195.
6. Hammernik, Kerstin, et al. "Learning a variational network for reconstruction of accelerated MRI data." Magnetic resonance in medicine 79.6 (2018): 3055-3071.
7. Professor John Pauly’s Lecture on “Reconstruction of Non-Cartesian Data”, Stanford University
8. Pruessmann, Klaas P., et al. "SENSE: sensitivity encoding for fast MRI." Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine 42.5 (1999): 952-962.

ISMRM & ISMRT ANNUAL MEETING & EXHIBITION

Honolulu, Hawai'i, USA
10-15 May 2025

STAY CONNECTED DURING THE MEETING:

ISMRM ISMRT.ISMRM | ISMRM ISMRT_ISMRM | ISMRM ISMRT-ISMRM | ISMRM_ISMRT | ISMRM ISMRT

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